

SHELL PROGRAMMING LAB GUIDE

Comparative Solutions for Linux Bash & Windows PowerShell

Question 1: Addition of Two Numbers

Write a shell script to read two numbers from the user and display their sum.

EXAMPLE EXECUTION

Input: 10, 20

Output: Sum = 30

Linux Bash Script (.sh)

```
#!/bin/bash
# Read the first number
read -p "Enter first number: " num1

# Read the second number
read -p "Enter second number: " num2

# Perform addition using standard arithmetic expansion
sum=$((num1 + num2))

# Display the result
echo "Sum = $sum"
```

Windows PowerShell Script (.ps1)

```
# Read the numbers and cast explicitly to integers
$num1 = [int](Read-Host "Enter first number")
$num2 = [int](Read-Host "Enter second number")

# Perform addition
$sum = $num1 + $num2

# Display the result
Write-Host "Sum = $sum"
```

Question 2: Even or Odd

Write a shell script to check whether a given number is even or odd.

EXAMPLE EXECUTION

Input: 25

Output: 25 is an Odd Number.

Linux Bash Script (.sh)

```
#!/bin/bash
read -p "Enter a number: " num

# Check remainder using the modulo operator (%)
if [ $((num % 2)) -eq 0 ]; then
    echo "$num is an Even Number."
else
    echo "$num is an Odd Number."
fi
```

Windows PowerShell Script (.ps1)

```
$num = [int](Read-Host "Enter a number")

# Use the -eq comparison operator for PowerShell conditional statement
if ($num % 2 -eq 0) {
    Write-Host "$num is an Even Number."
} else {
    Write-Host "$num is an Odd Number."
}
```

Question 3: Factorial of a Number

Write a shell script to find the factorial of a given positive integer.

EXAMPLE EXECUTION

Input: 5

Output: Factorial = 120

Linux Bash Script (.sh)

```
#!/bin/bash
read -p "Enter a positive integer: " num

original_num=$num
fact=1

# standard while loop with -gt operator
while [ $num -gt 1 ]; do
    fact=$((fact * num))
    num=$((num - 1))
done

echo "Factorial of $original_num = $fact"
```

Windows PowerShell Script (.ps1)

```
$num = [int](Read-Host "Enter a positive integer")

$original_num = $num
$fact = 1

# standard while loop using PowerShell syntax
while ($num -gt 1) {
    $fact = $fact * $num
    $num = $num - 1
}

Write-Host "Factorial of $original_num = $fact"
```

Question 4: Sum of Digits

Write a shell script to calculate the sum of all digits of a given number.

EXAMPLE EXECUTION

Input: 12345

Output: Sum of Digits = 15

Linux Bash Script (.sh)

```
#!/bin/bash
read -p "Enter a number: " num

sum=0

while [ $num -gt 0 ]; do
    digit=$((num % 10))    # Extract last digit
    sum=$((sum + digit))   # Add to sum
    num=$((num / 10))      # Integer division chops off last digit
done

echo "Sum of Digits = $sum"
```

Windows PowerShell Script (.ps1)

```
$num = [int](Read-Host "Enter a number")

$sum = 0

while ($num -gt 0) {
    $digit = $num % 10
    $sum = $sum + $digit
    # [Math]::Floor prevents PowerShell from generating floating point decimals
    $num = [Math]::Floor($num / 10)
}

Write-Host "Sum of Digits = $sum"
```

Question 5: Reverse a Number

Write a shell script to reverse the digits of a given number.

EXAMPLE EXECUTION

Input: 1234

Output: Reverse = 4321

Linux Bash Script (.sh)

```
#!/bin/bash
read -p "Enter a number: " num

reverse=0

while [ $num -gt 0 ]; do
    digit=$((num % 10))
    reverse=$(( (reverse * 10) + digit ))
    num=$((num / 10))
done

echo "Reverse = $reverse"
```

Windows PowerShell Script (.ps1)

```
$num = [int](Read-Host "Enter a number")

$reverse = 0

while ($num -gt 0) {
    $digit = $num % 10
    $reverse = ($reverse * 10) + $digit
    $num = [Math]::Floor($num / 10)
}

Write-Host "Reverse = $reverse"
```